

Smell with Genji: Rediscovering Human Perception through an Olfactory Game with AI

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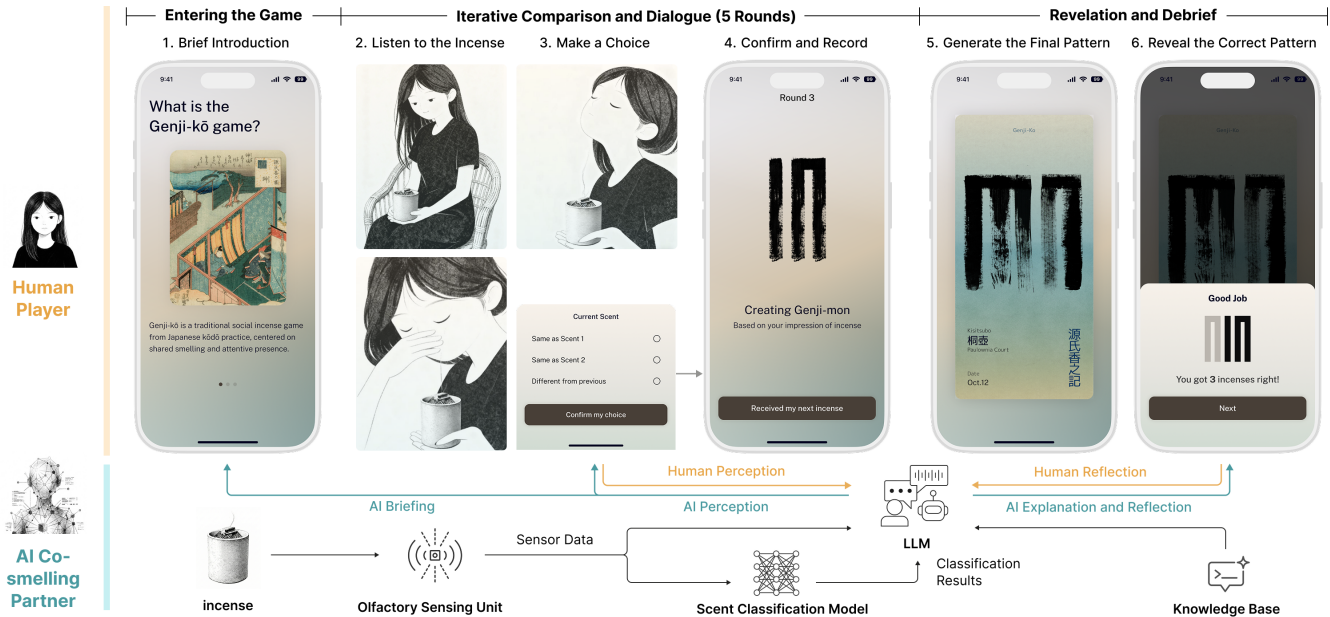


Figure 1: Diagram showing the interactive experience of *Smell with Genji*

Abstract

Olfaction plays an important role in human perception, yet its subjective and ephemeral nature makes it difficult to articulate, compare, and share across individuals. Traditional practices like the Japanese incense game *Genji-kō* offer one way to structure olfactory experience through shared interpretation. In this work, we present *Smell with Genji*, an artificial intelligence (AI)-mediated olfactory interaction system that reinterprets *Genji-kō* as a collaborative human-AI sensory experience. By integrating a game setup, a mobile application, and an AI co-smelling partner equipped with olfactory sensing and large language model (LLM)-based conversational capabilities, the system invites participants to compare scents and construct *Genji-mon* patterns, fostering reflection through a dialogue that highlights the alignment and discrepancies between human and machine perception. This work illustrates how sensing-enabled AI can participate in olfactory experience alongside users, pointing toward new possibilities for AI-supported sensory interaction and reflection in human-computer interaction (HCI).

CCS Concepts

• **Human-centered computing** → **Collaborative interaction**; **Natural language interfaces**; • **Applied computing** → **Arts and humanities**.

Keywords

Olfaction; Human-AI Interaction; Sensory Perception; Olfactory Games; Multisensory AI

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1 Introduction

Unlike vision or hearing, olfactory perception is closely linked to affective and memory-related processes, often bypassing high-level cognitive filtering [4, 7]. Prior work further suggests that awareness of smell is connected to how people experience mood and memory in everyday life, and that cultivating olfactory perception may influence aspects of well-being [1, 10]. However, this potential is difficult to leverage due to an olfactory-verbal gap [17]. Compared with other perceptual modalities, references to smell are infrequent across languages and communicative contexts, reflecting a limited

shared vocabulary for describing olfactory experience [2, 12–14]. As a result, while people generally have less difficulty reporting that they smell something, they often struggle to describe what they smell [9, 14]. In this paper, we explore this gap by introducing *Smell with Genji*, an AI-mediated experience that draws on a cultural ritual to support the cultivation of olfactory perception through guided comparison and reflection (Figure 1).

One way people have historically engaged with this challenge is through cultural practices that structure olfactory experience. In Japanese culture, *Genji-kō* is a social olfactory game rooted in *Kōdō*, the Way of Fragrance, in which participants compare sequences of five incense scents and communicate their perceptions [11, 19]. The game is relevant in this context because it emphasizes comparative olfactory judgment, allowing participants to attend to differences between scents rather than attempting to describe a single scent in isolation. Prior work such as eGenjiko digitized the hardware and matching mechanics of *Genji-kō*, but largely framed the experience as an individual interaction [19]. This approach overlooked the social dynamics central to *Genji-kō* and limited opportunities for deeper engagement and reflection around olfactory experience.

Recent advances in AI open new opportunities to extend such practices beyond digitization. LLMs have been widely adopted in HCI to support information analysis, interpretation, and interactive dialogue, enhancing engagement in learning and reflective interaction [16, 18]. However, LLMs do not directly perceive smell and therefore cannot engage with olfactory experience independently. In parallel, recent work in olfactory AI demonstrates that smells can be computationally represented through sensor-based signals, enabling AI systems to distinguish between different scent profiles [5]. Yet, much of this work has focused on detection, identification, or reproduction, with limited integration into interactive HCI contexts that support reflection and sense-making [6, 8]. Together, these developments suggest an opportunity to combine LLM-based interpretive and dialogic capabilities with olfactory sensing, enabling AI to engage more meaningfully within structured olfactory interactions.

Building on this opportunity, our work shifts attention from digitizing game mechanics toward augmenting participants' sensory awareness. We introduce *Smell with Genji*, a collaborative olfactory experience that builds on the *Genji-kō* framework by integrating an AI co-smelling partner into the interaction. The AI in our system interprets scents through temporal patterns in sensor signals, capturing changes in chemical responses over time. These sensor-derived representations are integrated with the LLM's capabilities for aggregation and dialogue. The contrast between human and machine perception creates moments of alignment and discrepancy that prompt reflective engagement with sensory judgments, helping participants move beyond vague impressions toward more deliberate articulation of olfactory experience [15]. Beyond human-AI interaction, the system also aggregates interpretations from prior sessions, enabling participants to ground their perceptions within a shared experiential space.

This work contributes an AI-mediated olfactory experience that extends an existing traditional olfactory game. Beyond language

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interpretation, the AI organizes sensory data, aggregates prior interpretations, and generates contextually grounded dialogue. Through this design, we demonstrate how a sensing-enabled AI can participate as a co-smelling partner within a culturally grounded interaction. We propose this system as a medium for cultivating sensory discovery, supporting reflection on human perception, and contributing to human well-being.

2 Interactive Experience

Smell with Genji is an AI-mediated olfactory interaction system built around the *Genji-kō* game. The system consists of a guided game setting with a mobile application and an AI co-smelling partner equipped with olfactory sensing and LLM-based conversational capabilities. Together, these components support a guided, multi-round experience in which participants engage with incense scents through comparison, reflection, and interaction, as shown in Figure 2.

Entering the Game: Through the mobile application and AI briefing, participants briefly familiarize themselves with the structure of the *Genji-kō* game and the role of *Genji-mon* patterns as a way of encoding similarities and differences across a sequence of scents. Before gameplay begins, participants enter a short calibration phase where five incense samples are presented sequentially. Participants are instructed to attend to each encounter as it unfolds. This phase establishes an olfactory baseline, supporting attention to subtle scent differences.

Iterative Comparison and Dialogue (5 Rounds): The core experience unfolds over five rounds, each following a consistent interaction loop. At the beginning of each round, participants “listen” to an incense, following on-screen guides that support breathing and focused attention. In the first round, this encounter establishes an initial reference and does not require a judgment. From the second round onward, participants indicate whether the current scent matches any of the previous ones or is distinct.

After making a selection, the AI co-smelling partner engages participants in a brief dialogue, asking about familiarity and perceived differences compared to earlier scents, while sharing its own interpretation. These interpretations are grounded in the AI’s sensing of the incense and may incorporate perspectives aggregated from prior participants. An example dialogue could be: “*In this rare alignment. We breathe as one human and machine. No edges, no shadows, just a shared scent suspended between memory and possibility.*” Participants can reflect on this exchange and may revise their choice before confirming their judgment.

Once the judgment is confirmed, the system programmatically updates the *Genji-mon* diagram in real time on the mobile application. Each round adds a vertical line representing the current incense, which is compared against all prior encounters. Scents perceived as the same are joined at the top, while distinct scents remain separated, progressively forming a *Genji-mon* pattern according to traditional *Genji-kō* rules. This loop-smelling, judging, dialogue, visualization-repeats across all five rounds. As the sequence progresses, participants continuously compare the current scent against multiple prior encounters, engaging memory, attention, and perceptual adjustment.

Throughout the interaction, participants encounter moments where their judgments align with or diverge from those of the AI. These points of productive tension invite closer reflection on how scent similarity is perceived and interpreted [3]. By comparing human intuition with machine-generated interpretations, the experience encourages participants to slow down, re-examine their assumptions, and attend more carefully to subtle differences between scents.

Revelation and Debrief: After completing the fifth round, the system reveals the full *Genji-mon* pattern generated through the participant’s choices, alongside a comparison with the correct pattern. Participants receive a bookmark featuring their final pattern, providing a tangible artifact that marks the completion of the game. The session concludes with a guided debrief where the AI explains how temporal patterns in sensor signals informed its judgments. This final “revelation” phase foregrounds the contrast between human and machine approaches to understanding smell. Through this comparison, participants are encouraged to reflect not only on the scents themselves, but also on their own approach for making sense of sensory experience.

3 System Architecture

Smell with Genji integrates time-series olfactory sensing with LLM-orchestrated dialogue and a mobile interaction layer to support a turn-based, socially grounded olfactory interaction.

3.1 Olfactory Game Setup

The mobile application (Figure 3a), implemented using React and WebSockets, serves as the coordination layer of the system. At session initialization, participants scan a QR code that associates the interaction with a predefined incense sequence and its correct *Genji-mon* pattern. This linkage allows the system to synchronize the physical incense with the system’s corresponding olfactory data. The mobile application manages round progression and decision input throughout the session. In each round, participants submit a comparative judgment indicating whether the current scent matches a previous one or is distinct. After confirmation, each decision is recorded and mapped to a programmatic implementation of the traditional *Genji-mon* rules. Through this process, the *Genji-mon* pattern is incrementally generated, forming a cumulative record of the participant’s perceptual judgments. Upon completion, the system renders the completed *Genji-mon* pattern alongside the correct pattern, supporting comparison and reflection.

At the center of the interaction space sits a custom-designed, 3D-printed enclosure that embodies the AI co-smelling partner (Figure 3b). Rather than serving as a literal representation of a “nose”, the device is designed to capture the ritual qualities of *Kōdō*. It houses the olfactory sensing components and a small display, serving as the AI’s physical presence within the game setting and anchoring human-AI interaction spatially during the ritual.

Participants engage with the olfactory experience using a hand-held incense holder consistent with traditional *Genji-kō* practice (Figure 3c). While the physical act of smelling is preserved, the surrounding game logic, notation, and coordination are digitally

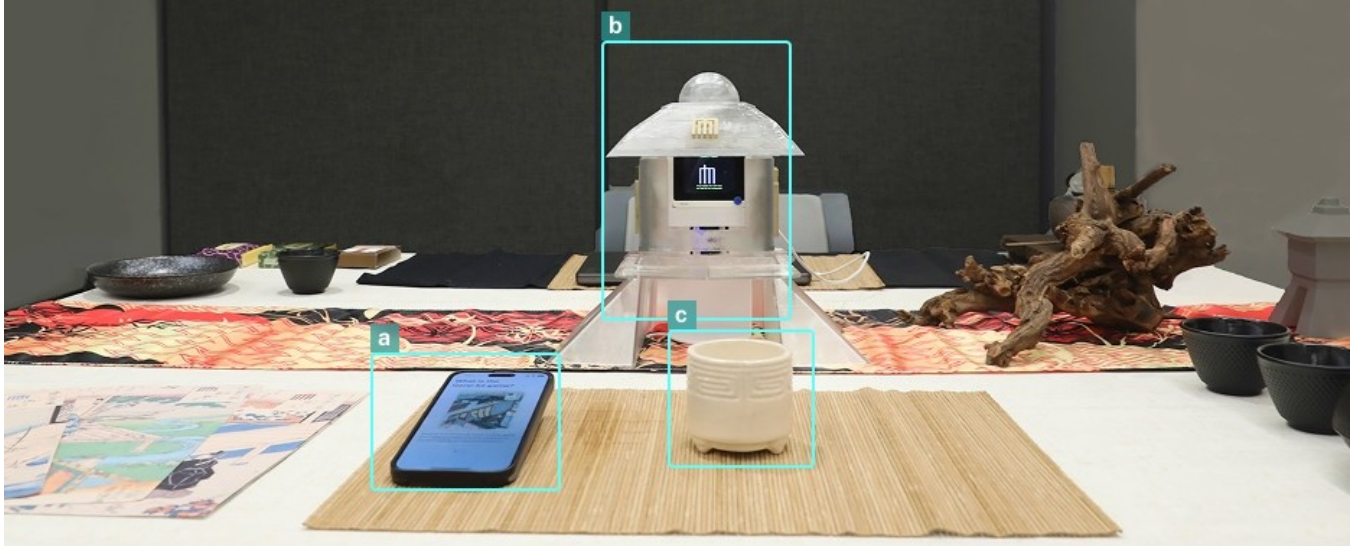


Figure 2: Overview of the *Smell with Genji* system. (a) Mobile application for guided gameplay and *Genji-mon* visualization. (b) AI co-smelling partner embodied in a 3D-printed enclosure with sensing and display. (c) Incense holder for smelling.

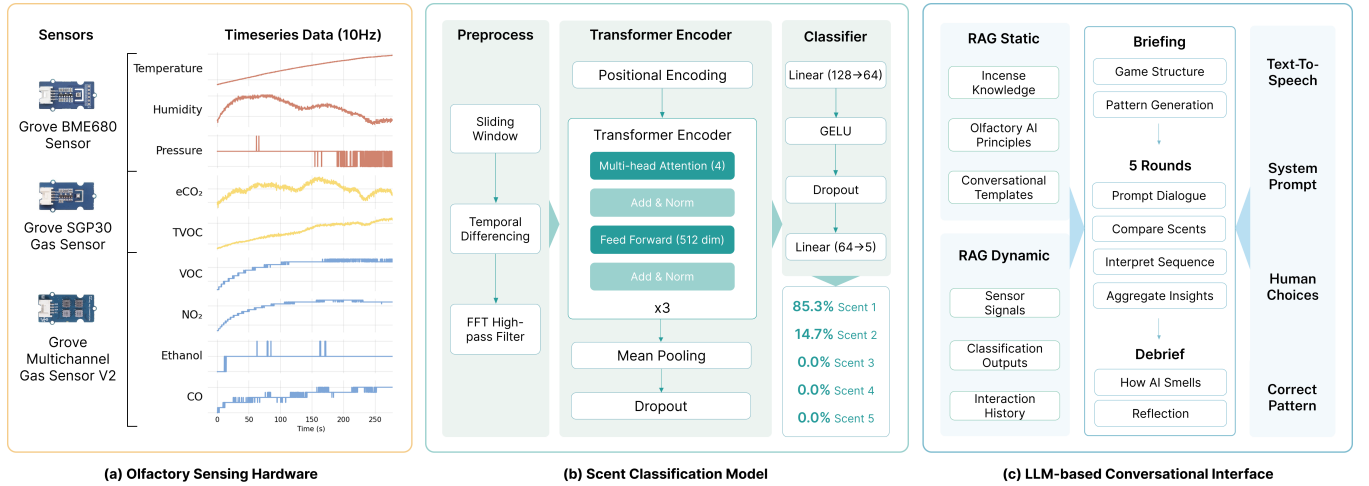


Figure 3: System Diagram of the AI Co-smelling Partner. (a) Olfactory sensing hardware: Three gas sensors collect time-series signals. (b) Scent classification model: A Transformer encoder processes signals to classify scents. (c) LLM-based conversational interface: A RAG-based pipeline integrates static knowledge and dynamic memory to facilitate the briefing, five-round interaction, and debriefing.

mediated through the mobile interface and the AI co-smelling partner to support guidance, recording, and reflection across the full interaction.

3.2 AI Co-smelling Partner

The AI co-smelling partner is implemented as a sensing-reasoning-dialogue pipeline embedded within the 3D-printed enclosure. It integrates olfactory sensing hardware, a scent classification model, and an LLM-based conversational interface to perceive, interpret, and communicate scent-related judgments alongside human participants (Figure 3).

Olfactory Sensing Hardware: The sensing system employed three sensors to capture environmental and gas composition data: Grove BME680 Sensor (temperature, humidity, pressure), a Grove SGP30 Gas Sensor (total volatile organic compounds (TVOC), equivalent carbon dioxide (eCO₂)), and a Multichannel Gas Sensor V2 (volatile organic compounds (VOC), nitrogen dioxide (NO₂), ethanol, and carbon monoxide (CO)). For each incense sample, 10 grams of material were placed in a 160 × 80 mm sealed plastic bag with sensors positioned 20 mm away. Incense was not burned to preserve sensor integrity. Data acquisition occurred at 10 Hz over

5-minute intervals, capturing 9 channels of sensor readings. Collection spanned varied environmental conditions (indoor/outdoor) and temporal settings (morning/evening), totaling 75 minutes of recording time and 405,000 data points.

Scent Classification Model: For the classification model, we experimented with temporal differencing, high-pass FFT filtering, standard scaling, and five window-stride configurations to optimize temporal feature extraction. The model architecture employs a Transformer encoder backbone coupled with an MLP classifier head for 5-class incense categorization. Under controlled environmental conditions, the model achieves approximately 40% accuracy due to overlapping VOC profiles from chemically similar incense samples. Our game compensates for this uncertainty by positioning the AI as a smell partner rather than a competitor, framing the experience as a shared journey where both the human and the AI are learning to smell. For real-time inference, we use windowed prediction with accumulative voting to generate prediction results for the game.

LLM-based Conversational Interface: This layer operates over two Retrieval-Augmented Generation (RAG) databases to support context-aware dialogue throughout the experience. A static database stores incense knowledge, olfactory AI principles, and conversational templates, while a dynamic database captures session-specific information, including sensor signals, classification outputs, and interaction history. Guided by a system prompt, the agent translates sensor-derived trends and model outputs into descriptive language and reflects on similarities and differences across the scent sequence throughout the five game rounds and during a final debriefing phase. Mode-specific RAG retrieval ensures contextually appropriate responses, while a custom LLM persona and text-to-speech synthesis support a consistent, voice-based co-smelling partner that facilitates reflective comparison.

4 Conclusion and Future Work

This work presents *Smell with Genji*, an AI-mediated olfactory interaction system that reinterprets the traditional *Genji-kō* game as a collaborative human-AI sensory experience while preserving its social dynamics. By integrating multichannel olfactory sensing, a pre-trained temporal scent classification pipeline, LLM-based reasoning, and a mobile interaction layer, the system enables AI to participate as a co-smelling partner. The AI perceives temporal scent dynamics, articulates interpretations grounded in both sensor data and shared human perspectives, and accompanies participants in reflective exploration. In doing so, *Smell with Genji* extends a culturally grounded practice into an interactive medium that supports comparison and dialogue around scent, offering a grounded way to engage with olfactory experience in HCI contexts where smell is often difficult to articulate or reflect upon.

Further user studies may inform refinements to the interaction design, including how dialogue, pacing, and visual cues support comparison and reflection. On the technical side, improved sensor configuration and temporal modeling could enhance sensitivity to subtle scent differences and enable more expressive AI interpretations. Beyond its current companion role, the AI co-smelling partner could adopt alternative personas, allowing investigation of how different conversational styles shape sensory awareness and reflection. More broadly, future work may examine how AI-mediated

olfactory interaction can support attentiveness and self-reflection, contributing to a deeper understanding of AI's role in augmenting human sensory experience.

References

- [1] Gregory N. Bratman, Cecilia Bembibre, Gretchen C. Daily, Richard L. Doty, Thomas Hummel, Lucia F. Jacobs, Peter H. Kahn, Connor Lashus, Asifa Majid, John D. Miller, Anna Oleszkiewicz, Hector Olvera-Alvarez, Valentina Parma, Anne M. Riederer, Nancy Long Sieber, Jonathan Williams, Jieling Xiao, Chia-Pin Yu, and John D. Spengler. 2024. Nature and human well-being: The olfactory pathway. *Science Advances* 10, 20 (2024), eadn3028. arXiv:https://www.science.org/doi/pdf/10.1126/sciadv.adn3028 doi:10.1126/sciadv.adn3028
- [2] Ewen L. Cameron, Michael R. Anderson, and Per Møller. 2016. The Accuracy, Consistency, and Speed of Odor and Picture Naming. *Chemosensory Perception* 9, 2 (jun 2016), 69–78. doi:10.1007/s12078-016-9204-4
- [3] Anna L. Cox, Sandy J.J. Gould, Marta E. Cecchinato, Ioanna Iacovides, and Ian Renfree. 2016. Design Frictions for Mindful Interactions: The Case for Microboundaries. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (San Jose, California, USA) (CHI EA '16). Association for Computing Machinery, New York, NY, USA, 1389–1397. doi:10.1145/2851581.2892410
- [4] Vanessa Danthiir, Richard D Roberts, Gerry Pallier, and Lazar Stankov. 2001. What the nose knows: olfaction and cognitive abilities. *Intelligence* 29, 4 (July 2001), 337–361. doi:10.1016/S0160-2896(01)00061-7
- [5] Dewei Feng, Carol Li, Wei Dai, and Paul Pu Liang. 2025. *SMELLNET: A large-scale dataset for real-world smell recognition*. arXiv:2506.00239 [cs.AI] https://arxiv.org/abs/2506.00239
- [6] Furizal Furizal, Alfian Ma'arif, Asno Firdaus, and Wahyu Rahmانيar. 2023. Future potential of E-Nose technology: A Review. *International Journal of Robotics and Control Systems* 3 (07 2023), 449–469. doi:10.31763/ijrcs.v3i3.1091
- [7] Rachel Herz and Trygg Engen. 1996. Odor memory: Review and analysis. *Psychonomic bulletin & review* 3 (09 1996), 300–313. doi:10.3758/BF03210754
- [8] Amanda K. Holloman and Chris S. Crawford. 2022. Defining Scents: A Systematic Literature Review of Olfactory-based Computing Systems. *ACM Trans. Multimedia Comput. Commun. Appl.* 18, 1, Article 15 (Jan. 2022), 22 pages. doi:10.1145/3470975
- [9] Andreas Keller and Leslie B. Vosshall. 2004. Human olfactory psychophysics. *Current Biology* 14 (2004), R875–R878. https://api.semanticscholar.org/CorpusID:5412859
- [10] C. Licon, C. Manesse, C. Roubay, and M. Bensafi. 2018. The sense of smell in relation to our affective states and wellbeing. In *The Routledge International Handbook of Psychobiology* (1 ed.), Philip N. Murphy (Ed.). Routledge, London, Chapter 4. doi:10.4324/9781315642765
- [11] Oran Looney. 2024. The Art and Mathematics of Genji-Kō. https://www.oranlooney.com/post/genji-ko/. Accessed: 2026-01-22.
- [12] Asifa Majid. 2020. Human olfaction at the intersection of language, culture, and biology. *Trends in Cognitive Sciences* 25 (12 2020). doi:10.1016/j.tics.2020.11.005
- [13] Asifa Majid and Niclas Burenholt. 2014. Odors are expressible in language, as long as you speak the right language. *Cognition* 130, 2 (2014), 266–270. doi:10.1016/j.cognition.2013.11.004
- [14] Jonas Olofsson and Jay Gottfried. 2015. The muted sense: Neurocognitive limitations of olfactory language. *Trends in Cognitive Sciences* 19 (05 2015). doi:10.1016/j.tics.2015.04.007
- [15] Corina Sas and Alan Dix. 2009. Designing for reflection on experience. 4741–4744. doi:10.1145/1520340.1520730
- [16] Orit Shaer, Angelora Cooper, Osnat Mokryn, Andrew L. Kun, and Hagit Ben Shoshan. 2024. AI-augmented brainwriting: Investigating the use of LLMs in group ideation. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 1050, 17 pages. doi:10.1145/3613904.3642414
- [17] Richard J. Stevenson and Donald A. Wilson. 2007. Odour perception: An object-recognition approach. *Perception* 36, 12 (2007), 1821–1833. doi:10.1068/p5563
- [18] Diyi Yang. 2024. Human-AI Interaction in the Age of Large Language Models. *Proceedings of the AAAI Symposium Series* 3, 1 (May 2024), 66–67. doi:10.1609/aaais.v3i1.31183
- [19] Anna Yokokubo, Mudmee Chaichirawiwat, Guillaume Lopez, Yasushi Matoba, Miku Ohno, and Itiro Sii. 2019. eGenjiko: Scent Matching Game using a Computer-Controlled Censer. In *Extended Abstracts of the Annual Symposium on Computer-Human Interaction in Play Companion Extended Abstracts* (Barcelona, Spain) (CHI PLAY '19 Extended Abstracts). Association for Computing Machinery, New York, NY, USA, 789–795. doi:10.1145/3341215.3356259